different style of testing, where the test is verifying the behavior of the system during the test rather than the state of the system at the end of the test.

Test doubles can be a bit of a contentious issue, with different people giving conflicting advice about the best way to use them when testing. I'd like to give you enough information to make informed decisions about who to agree with. (Though I hope you'll agree with me.) I'll start by looking at the mechanics of test doubles, using the rspec-mocks library. Then I'll discuss different ways to use doubles to solve both of the testing issues described at the start of this chapter. Later, in Chapter 12, *Minitest*, on page 241, I'll cover the basics of the Mocha mocking library for use with Minitest.

Test Doubles Defined

One complicating factor in dealing with test doubles is that pretty much everybody who creates a tool feels perfectly free to use slightly different naming conventions than everybody else. Here are the names I use, which are—of course—also the correct ones. (This naming structure is the creation of Gerard Meszaros in *xUnit Test Patterns [Mes07]*.)

The generic term for any object used as a stand-in for another object is *test double*, by analogy to "stunt double" and with the same connotation of a cheaper or more specialized replacement for a more expensive real object. Colloquially, *mock object* is also sometimes used as a generic term but, confusingly, is also the name of a specific type of test double.

A *fake* is typically an ordinary Ruby object that you define that is designed to be used only in tests. The fake matches the API of an object but has vastly simplified or canned internals. Fakes are not typically created using a test-double library, but might be custom-created for use in a specific context. A good example is the Fake Stripe gem,[1] which creates a fake instance of the Stripe server to intercept API calls in a test situation.

A *stub* is a fake object that returns a predetermined value for a method call without calling the actual method on an actual object. A stub can be a new object created only to be a stub, which you can create in RSpec using the `double` method. Alternatively, you can create a partial stub by stubbing a specific method on an existing object:

```
allow(thing).to receive(:name).and_return("Fred")
```

1. https://github.com/thoughtbot/fake_stripe